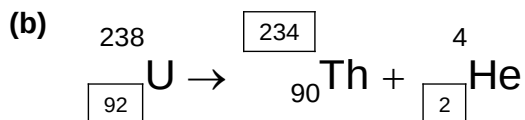


- 8 (a) The number 238 refers to the number of nucleons in the nucleus or number of protons and neutrons in the nucleus.



(c) (i) kinetic energy = $40.0 \times (5.90 \times 10^3)(2.70 \times 10^{-18})$
 $= 6.372 \times 10^{-13}$
 $= 6.37 \times 10^{-13} \text{ J (shown)}$

(ii)

$$\text{kinetic energy, } E_k = \frac{1}{2}mv^2 = \frac{1}{2} \frac{(mv)^2}{m} = \frac{p^2}{2m}$$

Since uranium nucleus was initially at rest, by the principle of conservation of momentum,

$$p_{Th} + p_{\alpha} = 0$$

$$p_{Th} = -p_{\alpha}$$

$$= -\sqrt{2m_{\alpha}E_{\alpha}}$$

$$= -\sqrt{2 \times (4 \times 1.66 \times 10^{-27})(6.37 \times 10^{-13})}$$

$$= -9.1975 \times 10^{-20} = -9.20 \times 10^{-20} \text{ N s}$$

(iii)

$$E_{k,Th} = \frac{p_{Th}^2}{2m_{Th}} = \frac{(9.20 \times 10^{-20})^2}{2(234 \times 1.66 \times 10^{-27})} = 1.0895 \times 10^{-14} = 1.09 \times 10^{-14} \text{ J}$$

$$E_{k,Total} = E_{k,Th} + E_{k,\alpha}$$

$$= (1.09 \times 10^{-14}) + (6.37 \times 10^{-13})$$

$$= 6.479 \times 10^{-13} \text{ J}$$

$$= \frac{6.479 \times 10^{-13}}{(10^6)(1.60 \times 10^{-19})} \text{ MeV}$$

$$= 4.0494 = 4.05 \text{ MeV}$$

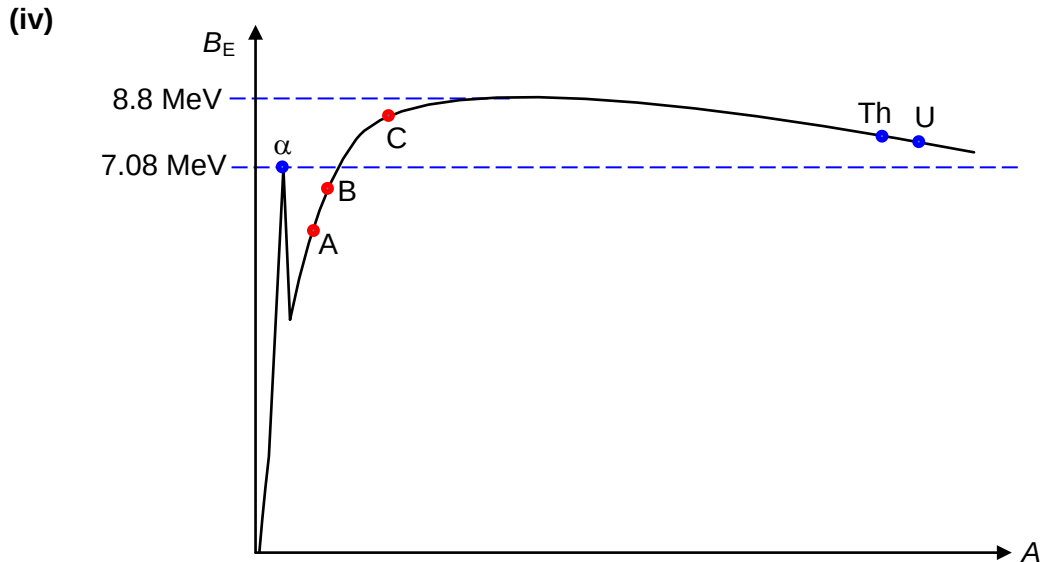
- (iv) No other particles or photons were emitted during the decay.

- (d) (i) Nuclear binding energy is the energy equivalent of the mass defect of a nucleus. It is the energy required to separate to infinity all the nucleons of a nucleus.

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- (ii) Possible nuclides: iron-56 (^{56}Fe), iron-58 (^{58}Fe), nickel-62 (^{62}Ni)
 $B_E = 8.8 \text{ MeV}$

(iii) $\Delta E = E_{k,\text{Total}} = BE_{Th} + BE_{\alpha} - BE_U$
 $4.05 = BE_{Th} + (4 \times 7.08) - (238 \times 7.57)$
 $BE_{Th} = 1777.39 \text{ MeV}$
 $\frac{BE_{Th}}{A_{Th}} = \frac{1777.39}{234}$
 $= 7.5957 = 7.60 \text{ MeV}$



On the left part of curve, He or α -particle on the sharp peak.

On the right part of curve, U (7.57 MeV) above α (7.08 MeV), and Th (7.60 MeV) just above U and just to its left.

- (v) 1. Nuclear fusion occurs when two light nuclei combine to form a nucleus of greater mass.
2. All A, B and C on the left of the highest B_E point. Nuclei A and B lighter than or on the left of C, but B_E of C is higher than both nuclei A and B.